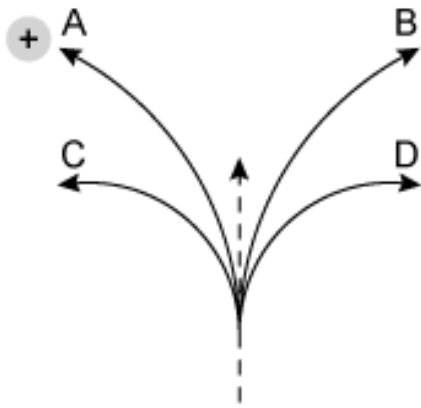


In the presence of an external magnetic field (not shown), a proton initially traveling upward is deflected into the path labeled as A in the following figure.



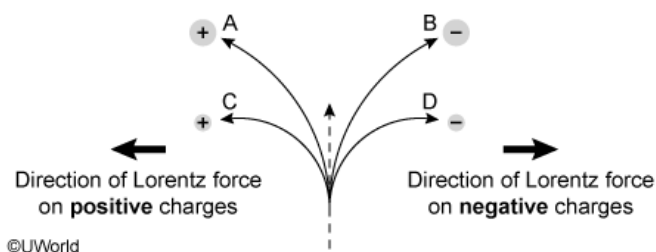
An electron with the same initial velocity as the proton travels through the magnetic field. What would be the expected path of the electron?

- ☐ A. A [10%]
- ☐ B. B [60%]
- ☐ C. C [4%]
- ✓ ☒ D. D [24%]

Correct

24% answered correctly

Explanation:



Particles with charge q and speed v are accelerated in a magnetic field B by the **Lorentz force** F , which has magnitude:

$$F = qvB$$

The direction of the force is determined by using the **right-hand rule** and is always perpendicular to the particle's velocity. As a result, the particle's trajectory is forced into a curved path.

In the figure, the path of the proton is bent to the left in the presence of the

external magnetic field. Consequently, the Lorentz force initially points **to the left for positive charges** in the magnetic field. However, because an electron is negatively charged, the trajectory of an electron would curve in the opposite direction, which is **toward the right**.

Although the Lorentz forces on the proton and electron have identical magnitudes, their paths are determined by the acceleration a imparted on each particle. **Newton's second law of motion** implies that acceleration is **inversely proportional** to particle mass m :

$$a = \frac{F}{m} = \frac{qvB}{m}$$

Because the **mass and inertia** of an electron are much smaller than those of a proton, the electron experiences greater acceleration. Therefore, its **path will have a greater curvature (Choice D)**.

(Choices A and C) The trajectory of an electron would curve in the opposite direction (ie, toward the right).

(Choice B) A particle with a negative charge and the mass of a proton would follow this path.

Educational objective:

The Lorentz force exerted on a moving charge in the presence of a magnetic field is perpendicular to the particle's velocity. As a result, the charge's trajectory is forced into a curved path, with negative charges and positive charges bent in opposite directions. Because an electron is less massive than a proton, an electron's path would be more curved in the same magnetic field.

Time spent:0 sec

Subject:
Physics

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System:
Electrostatics and Circuits